Hritabrata Das

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Education

Chitkara University — Rajpura, Punjab, India
B.E. Computer Science and Engineering – AI & ML Specialization | ***CGPA: 9.49 / 10*** — *Aug. 2024 – May 2028*

The G.V.E.A. — Balurghat, West Bengal, India
Primary – High School (CISCE Curriculum) | ***86%*** | *Spoken English, Competitive Debating* — *Jun. 2008 – Mar. 2024*

Experience and Achievements

Head of Web Development — Sept. 2025 – Present
Google Developer Groups (GDG) On Campus – Chitkara University (promoted from Web Developer) — *Rajpura, Punjab, India*

- Lead the web development team for GDG On Campus Chitkara, owning technical direction for the chapter's web platforms and a community of 150+ student developers
- Built and shipped 2+ web platforms for the chapter and led full-stack development for 3+ community events and workshops, increasing student participation in Google tech programs by 40%
- Mentored team members on web development best practices, conducting 3+ internal code review sessions per semester

- Secured top-3 finishes in 4+ intra-college competitions across coding, design, and aptitude tracks
- Placed **3rd** among 20+ teams in the **Agentic Sprint Hackathon** (on-campus), leading a team that delivered a functional autonomous AI agent within a 24-hour constraint

Projects

R.I.S.H.I. | *Electron, Qdrant, RAG, whisper, OpenCLaw, EC2, Python* | Code — March. 2026 – Present

- Developed an autonomous AI companion executing real tasks end-to-end across devices — web search, image generation, file management, and email — through a unified backend
- Engineered multi-provider LLM routing across 6+ models (Claude, DeepSeek, Llama, Mixtral, Azure, Gemini, GitHub Copilot, Nvidia NIM) with long-term memory via RAG over a Qdrant vector DB
- Deployed core intelligence on AWS EC2 via an OpenClaw middleware layer, with GPT-4o as the primary reasoning brain and Gemini handling web search and embeddings

SWEPilot | *Azure B1s VM, Azure AI Foundry, GitHub REST API, FastAPI* | Code | Live — Sept. 2026 – Present

- Built an autonomous AI software-engineering agent that turns GitHub issues into tested, review-ready Pull Requests, combining mini-SWE-agent with an Azure-hosted LLM to analyze issues, edit code, and run tests
- Designed a deterministic orchestration layer separating AI-driven coding from workflow control: webhook-triggered issue detection, isolated Azure VM workspaces, automated branch/commit/push, and GitHub REST API PR creation
- Architected the workflow to keep humans in the loop by design, every AI-generated change lands as a reviewable Pull Request rather than a direct commit to the production branch

Exposr | *Python, asyncio, TCP/UDP, Azure* | Code | Live — September. 2026 – Present

- Built and published a reverse TCP/UDP tunneling CLI (PyPI, v0.5.4) that exposes local services publicly through a persistent-agent relay, tracking each connection with a dedicated UUID-tunneled data channel
- Engineered dynamic public port registration across a 10,000-port fallback range (20000–30000) with automatic agent reconnection, supporting unlimited simultaneous public connections
- Benchmarked relay performance on Azure across 100 sequential requests, isolating ˜435ms of protocol overhead from network RTT and identifying an 85% throughput drop (12.4 → 1.8 req/s) as the target for a planned connection-reuse redesign

Technical Skills

Languages: Python, JavaScript, TypeScript, C++, Java, SQL (PostgreSQL), HTML, CSS
Frameworks & Libraries: React, Next.js, React Native, Node.js, Express.js, FastAPI, Django, Flask, Tailwind CSS, NumPy, Pandas, Matplotlib, Scikit-Learn
AI & ML: RAG (Retrieval-Augmented Generation), Vector Databases (Qdrant), LLM Integration (OpenRouter, Azure, Gemini), faster-whisper, Piper TTS, Agentic AI Systems
Tools & Cloud: Git, GitHub, Docker, Redis, Twilio, Razorpay, REST API Development, Oracle Cloud (ARM), Microsoft Azure (B1/B1s), AWS EC2, Google Cloud Platform

Certifications

Microsoft AI Fundamentals | Microsoft Azure Fundamentals | Microsoft Azure Data Fundamentals |
Microsoft Power BI Data Analyst Associate | Design Thinking – University of Virginia | Introduction to Cyber Security